



## Research



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# New high-resolution subglacial bed topography and coastal bathymetry of the Antarctic and Greenland Ice Sheets

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The description of the bed topography under the Greenland and Antarctic Ice Sheets has greatly improved over the past decade through new field campaigns and mapping techniques, leading to BedMachine, a high-resolution gridded bed map widely used by the ice-sheet modelling community. Despite regular updates, BedMachine still suffers from uncertainty in ocean bathymetry and mapping artefacts in the ice-sheet interior. We describe here four recent improvements that address these limitations. In Greenland, we use ICESat-2 surface elevation time series to construct an ensemble of bed elevations that captures finer bed details. In Antarctica, we use Ice-Flow Perturbation Analysis in the interior. This approach provides an estimate of the bed topography

using the surface expression of mesoscale bedforms. For periphery ice caps and the Antarctic Peninsula, we use the machine learning-based IceBoost approach, which is capable of inferring fine details based on surface features. Finally, over the continental shelf, we use a new gravity inversion product from the Antarctic Gravity Anomaly Grid, which provides significant refinements to the bathymetry around the entire ice sheet. Overall, these represent major improvements in the description of the bed topography and bathymetry for both ice sheets.

This article is part of the Theo Murphy meeting issue 'Next generation ice-sheet bed measurements'.

## 1. Introduction

The topography under the Antarctic and Greenland Ice Sheets, as well as the ocean bathymetry along their periphery, exerts a primary control on their current and future vulnerability to climate change. The shape of the bed in grounding zones, for example, controls the flux and therefore the ice discharge into the ocean, thereby influencing the stability of the grounding line [1,2]. Bed topography also governs the pathways through which warm ocean water reaches ice shelves [3–5] and where subglacial meltwater flows, affecting both basal melt rates and basal effective pressure [6]. Moreover, it influences the location of pinning points that provide buttressing support to ice shelves, the loss of which can lead to rapid inland ice acceleration [7]. Together, these factors make subglacial topography a key determinant of how ice sheets respond to climate forcing.

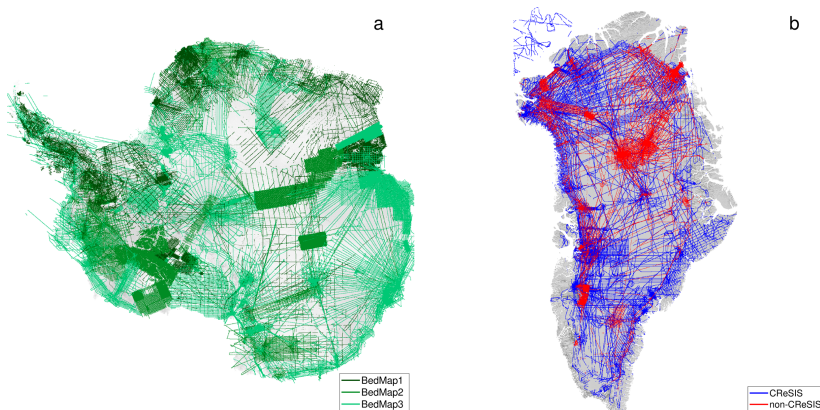
Bed topography is most efficiently measured by airborne ice-penetrating radar [8]. Over the past two decades, through the acquisition of new airborne-radar surveys [9] (figure 1) and the development of new mapping techniques, such as the conservation of mass [10,11], significant improvements in our knowledge of the subglacial topography of the Greenland and Antarctic Ice Sheets have been made. The development of BedMachine Greenland [12], BedMachine Antarctica [13] and Bedmap3 [14] profoundly improved our ability to model ice sheet flow [15]. We focus here specifically on BedMachine, which is a high-resolution (150 m for Greenland, 500 m for Antarctica) bed topography map, accompanied by an ice mask and surface elevation map that are all self-consistent. While Bedmap3 uses a simple interpolation approach, based on adapted thin plate splines [14], BedMachine relies on the principle of conservation of mass in fast-flowing regions, which reveals details in the bed topography in regions where radar measurements are not available [12,13]. BedMachine is regularly updated as new ice thickness measurements are integrated into the dataset. Despite these major improvements, however, uncertainties remain in many critical regions.

BedMachine Greenland v5, BedMachine Antarctica v3 and Bedmap3 still suffer from large uncertainty in ocean bathymetry over the continental shelf, and mapping artefacts in the interior of the ice sheet, where mass conservation cannot be used because of uncertainty in InSAR-derived velocity directions and magnitude. In this paper, we describe four recent developments that are integrated into the new versions of BedMachine and that help address the current limitations.

First, in Greenland, instead of relying on a single estimation of the bed through the use of mass conservation, we use the unprecedented series of ice elevation measurements from ICESat-2 combined with time series of ice flow velocities from NASA's MEaSUREs initiative, and two high-resolution surface mass balance reconstructions, to reconstruct an ensemble of bed elevations. We apply this approach to reassess estimates of and uncertainties in bed elevation at marine-terminating glaciers along the margin of the Greenland Ice Sheet.

Second, in Antarctica, we rely on Ice-Flow Perturbation Analysis (IFPA) [16] over grounded ice, where mass conservation is not possible. This approach provides an estimate of the bed topography using the surface expression of mesoscale (2–50 km) bedforms.

Third, for smaller ice caps around the periphery of both ice sheets, instead of relying on a simple shallow-ice-based approach [17], we use the machine learning model IceBoost [18]. This



**Figure 1.** Ice-penetrating radar measurements available for the Antarctic (*a*) and Greenland (*b*) ice sheets currently included in BedMachine. CRaSIS is the Center for Remote Sensing and Integrated Systems and processed most of Greenland's radar data.

approach uses several surface features (e.g. surface elevation, slope, curvature, surface speed and surface mass balance) to infer the bed topography and is able to capture more details than other existing approaches.

Lastly, over the continental shelf of Antarctica, we use a new inversion product from the Antarctic Gravity Anomaly Grid (AntGG [19]), which provides significant refinements to the bathymetry around the entire ice sheet.

## 2. Method

We provide here short descriptions for the four mapping techniques that are now included in BedMachine Greenland v6 and BedMachine Antarctica v4. For more detail, we refer the reader to the respective papers, where each method is described in ample detail.

### (a) Ensemble based on ICESat-2 time series

This is a new approach that we apply to the coast of Greenland. Instead of relying on a single bed topography inferred through mass conservation, we use time series of observations from ICESat-2 [20] to infer an ensemble of bed topography realizations.

Specifically, the mass conservation approach requires solving the following problem:

$$\begin{cases} \nabla \cdot H\bar{\mathbf{v}} = \dot{a} & \text{in } \Omega \\ H = H_{obs} & \text{on } \Gamma_- \end{cases}, \quad (2.1)$$

where  $\Omega$  is the model domain,  $\Gamma_-$  is the inflow boundary,  $\bar{\mathbf{v}}$  is the depth-averaged velocity,  $\dot{a} = \text{SMB} + \text{BMB} - \partial H / \partial t$  is the apparent mass balance, with SMB the surface mass balance, BMB the basal mass balance and  $H$  the ice thickness.

To account for all measurements of ice thickness,  $H_{obs}$ , along flight tracks,  $T$ , that lie within the model domain  $\Omega$ , we formulate an optimization problem [10,21], where we minimize the following cost function:

$$\mathcal{J}(H) = \int_T \frac{1}{2} (H - H_{obs})^2 dT + \int_{\Omega} \gamma_{\parallel} (\nabla H \cdot \mathbf{n}_{\parallel})^2 d\Omega + \int_{\Omega} \gamma_{\perp} (\nabla H \cdot \mathbf{n}_{\perp})^2 d\Omega, \quad (2.2)$$

where  $\mathbf{n}_{\parallel}$  and  $\mathbf{n}_{\perp}$  are unit vectors parallel and perpendicular, respectively, to the ice velocity, and  $\gamma_{\parallel}$  and  $\gamma_{\perp}$  are constant regularization parameters.

**Table 1.** Statistics and datasets used for bed realizations in previous versions of BedMachine Greenland versus new version. ATL is short for advanced topographic laser altimeter system.

| input                                      | BedMachine Greenland v5     | BedMachine Greenland v6 |
|--------------------------------------------|-----------------------------|-------------------------|
| ice elevation                              | GIMP [22]                   | ICESat-2 (ATL 14)       |
| thickening rate<br>$\partial H/\partial t$ | Smith <i>et al.</i> [20]    | ICESat-2 (ATL 15)       |
| surface mass balance (SMB)                 | RACMO 2.3 [23]              | RACMO and MAR           |
| ice velocity                               | Mouginot <i>et al.</i> [24] | MEaSURES                |
| number of bed realizations                 | 1                           | > 24                    |

Solving this optimization problem for one point in time, which is the current approach of BedMachine, provides one single bed estimate. Instead, we construct an ensemble of 24 bed reconstructions combining one bed topography for each of the 12 ICESat-2 cycles, every 90 days, and one of two different surface mass balance models (Modèle Atmosphérique Régional (MAR) and the Regional Atmospheric Climate Model (RACMO); see table 1). All these realizations should converge toward the ‘true’ bed topography and allow us to better calculate the uncertainty in mass-conservation inferred bed topography. We then use an ensemble mean or median for BedMachine, as it has been shown that ensemble means perform significantly better than single estimates [25,26].

(b) Ice-flow perturbation analysis

In the interior of the Antarctic ice sheet, where ice velocities are too slow to apply mass conservation ( $<30\text{ m a}^{-1}$ ), we employ the IFPA [16]. IFPA is based on the transfer of basal properties to the surface of the ice sheet by flowing ice [27]. Specifically, it utilizes a mathematical representation of the observed relationship between patterns in surface elevation and velocity and variations in bed topography and basal slipperiness beneath the ice sheet. This method leverages the recent availability of high-resolution satellite observations of ice-surface elevation and velocity over the entire ice sheet to provide spatially uniform information about mesoscale (2–50 km) bedforms. As IFPA infers perturbations to the long-wavelength topography and requires regional mean elevations from geophysical survey data, we calculate errors based on distance to survey points following the same methodology previously used for streamline diffusion [13]. This improved representation of topographic roughness is a significant advance from previous topographic maps, in which roughness has been artificially correlated with geophysical survey locations. IFPA is particularly valuable in areas where direct measurements of the ice-bed interface are sparse and where mass conservation is difficult to apply in the interior of the ice sheet.

(c) IceBoost for periphery glaciers

To better capture the topography of Greenland and Antarctic periphery glaciers, as well as the Antarctic Peninsula, we rely on the new machine learning system IceBoost v. 1.1 [18]. IceBoost employs an ensemble of two gradient-boosted decision trees trained globally on available ice-penetrating radar thickness data. The model utilizes 39 input features, including geodetic information, mass balance and ice velocity, to predict ice thickness distributions (see table 1 of [18] for a complete list). It has been shown that this model performs particularly well over high-latitude glaciers, reducing errors by up to 30–40% compared with existing models [18]. A key advantage of IceBoost is its ability to generalize predictions beyond the training domain. For this study, the model was retrained using an extended training global dataset that includes an additional

2 868 000 ice thickness measurements collected over the Antarctic Peninsula sourced from the Center for Remote Sensing and Integrated Systems (CReSIS). The data were manually quality-checked and outliers were removed as much as possible by assessing the consistency of crossover values at intersecting tracks. A cross-validation routine carried out on ground-truth data from the extended datasets in Antarctica and Greenland indicates IceBoost errors of 155 and 83 m, respectively, compared with 171 m for BedMachine Antarctica v3 and 106 m in Greenland v5.

#### (d) Antarctic Gravity Anomaly Grid gravity inversions

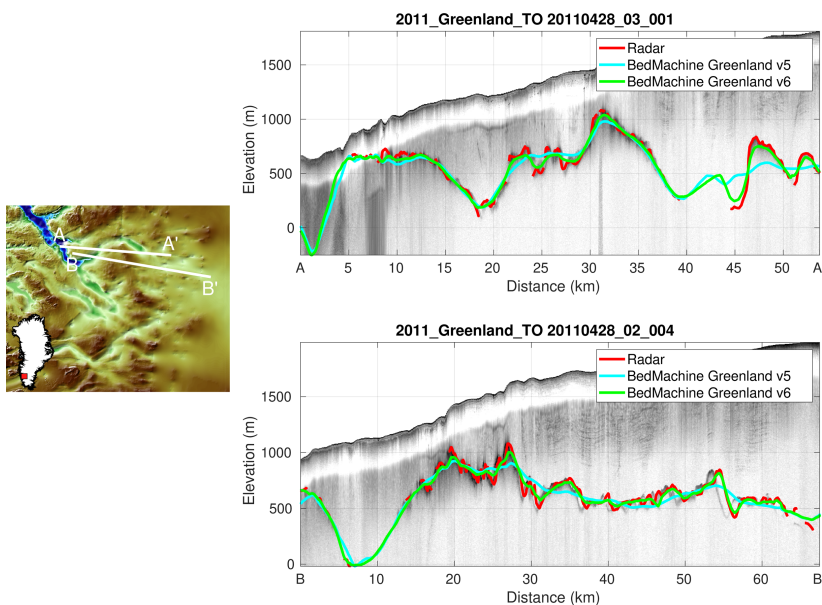
To improve the bathymetry along the periphery of the Antarctic Ice Sheet, we use a new gravity-based three-dimensional inversion recently released [19]. Inversions were calculated for all Antarctic ice shelf cavities and extended to the continental shelf. This dataset builds on the AntGG2021 gravity anomaly dataset [28], a refined 5 km resolution grid compiled from terrestrial, airborne and satellite-based gravity measurements. The authors used a layered Earth model comprising ice, water and bedrock with realistic density assumptions and constrained the gravity inversion using known seafloor depth observations from multibeam echo sounding (MBES), seismic surveys, autonomous underwater vehicles and data from marine mammals. Computed gravity anomalies from an initial three-layer model were iteratively compared with the ANTGG2021 gravity anomaly grid. The best estimate of bathymetry is found when the misfit between observed and calculated bathymetry reaches the 0.1 mGal threshold. The precision of inverted bathymetry is 27 m, with an estimated accuracy of 80 m for areas where MBES measurements are dense. Intermediate accuracy is found in areas with poor *in situ* data, reaching up to 150 m. Large uncertainties in the open-air gravity grid affect the accuracy of the inverted bathymetry, with maximum uncertainties reaching 250 m between Victoria Land and the Queen Mary coast, where no *in situ* data were available. This new dataset revealed previously unmapped deep troughs and cavities critical for understanding ice–ocean interactions and vulnerability to climate change.

### 3. Results and discussion

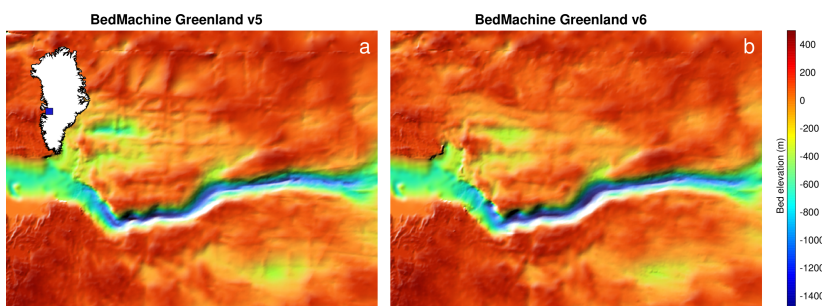
First, we present the results of the ensemble-based mapping approach of coastal Greenland. Figure 2 shows an example of application for Kangiata Nunata Sermia in West Greenland. We find that the ensemble median is overall in better agreement with the radar-inferred bed than the BedMachine Greenland v5. For this region, we find that the mean absolute difference between the inferred bed elevation and the radar data decreases from 66 to 44 m, which is close to the radar uncertainty. We also find that the current version of BedMachine tends to be smoother (due to the regularizing term of equation (2.2)) than the ensemble-based approach that better captures the roughness of the bed, as apparent in the radar data.

In some regions, as illustrated in figure 3 for Sermeq Kujalleq (or Jakobshavn Isbræ), West Greenland, we find that the flight lines were visible in the final topography of the current version of BedMachine, due to overfitting. Although these artefacts are not very common in BedMachine, they do create artificial ridges along flight lines that could impact numerical simulations of ice sheet flow by providing false anchor points. We see that the ensemble-based map has fewer of these artefacts (figure 3), suggesting that the ensemble-based approach is therefore less prone to overfitting. Overall, the ensemble-based approach leads to a reduced uncertainty in bed topography, with fewer mapping artefacts and roughness that is better captured.

In Antarctica, the vast majority of the ice flows slowly ( $<30 \text{ m a}^{-1}$ ), which makes mass conservation impossible in the interior of the ice sheet. In previous versions of BedMachine Antarctica, the interior was mapped using a simple streamline diffusion approach [13]. Instead, IFPA employs the physics of ice flow to infer the topography of the bed based on the observed ice-surface elevation and velocity [16]. Figures 4 and 5 show the current and new bed topography of two sectors of the ice sheet: the Roi Baudouin ice shelf and Thwaites Glacier.

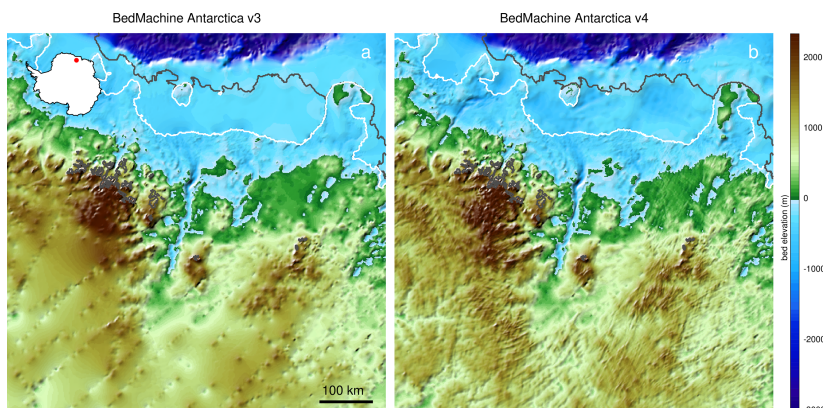


**Figure 2.** Kangiata Nunata Sermia; left: bed topography from BedMachine Greenland v6, with two flight lines (AA', BB') shown in white. The corresponding echograms are shown on the right, with the radar-picked bed topography in red, BedMachine v5 in cyan and BedMachine v6 in green.

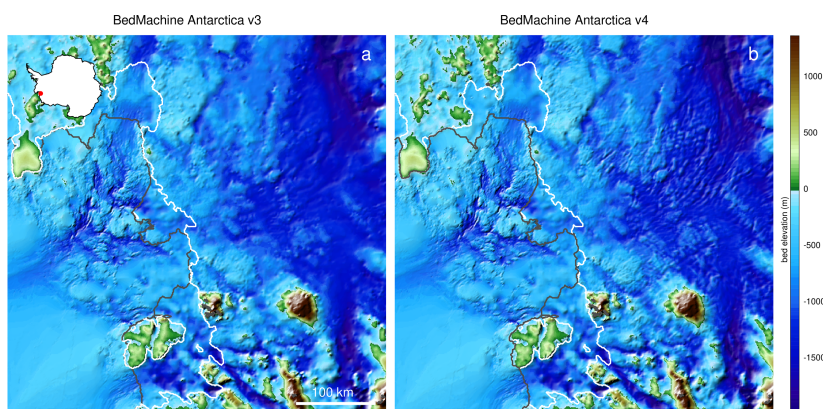


**Figure 3.** Bed topography of Sermeq Kujalleq (Jakobshavn Isbræ) from BedMachine Greenland v5 (a) and BedMachine Greenland v6 (b) in shaded relief, colour coded between  $-1500$  and  $+510$  m. Overfitting is apparent along the northern side of the glacier in (a) where a 'grid-like' pattern emerges.

Along the southern half of figure 4, we clearly see the typical artefacts of streamline diffusion where flight lines are apparent together with a smooth bed topography in between flight lines. This topography is in contrast to the inference from IFPA, where the topography matches with radar lines, and the basal roughness is spatially consistent across the region. Thwaites Glacier has been extensively surveyed, and mapping artefacts are not as clearly visible in figure 5. However, IFPA indicates again a rougher topography in the interior of the ice sheet. We should note that, while the majority of topographic features revealed by IFPA are indeed due to a transfer of basal features to the surface, there are a few isolated cases in the interior where surface processes occur, such as megadunes in East Antarctica [29], meaning that the ice surface which is inverted is not entirely representative of perturbations in the bed topography. IFPA cannot make this distinction, and so some of the bumps and ridges may be artefacts of this mapping method. In some regions, however, we have explored the use of different surface datasets to distinguish consistent features from transient features caused by artefacts in the surface datasets. Changing from the ITS\_LIVE



**Figure 4.** Bed topography from BedMachine Antarctica v3 (a) and v4 (b) in the area of the Roi Baudouin Ice Shelf, East Antarctica. The white thin line shows the location of the grounding line, and the grey line shows the position of the edge of the floating ice shelf.

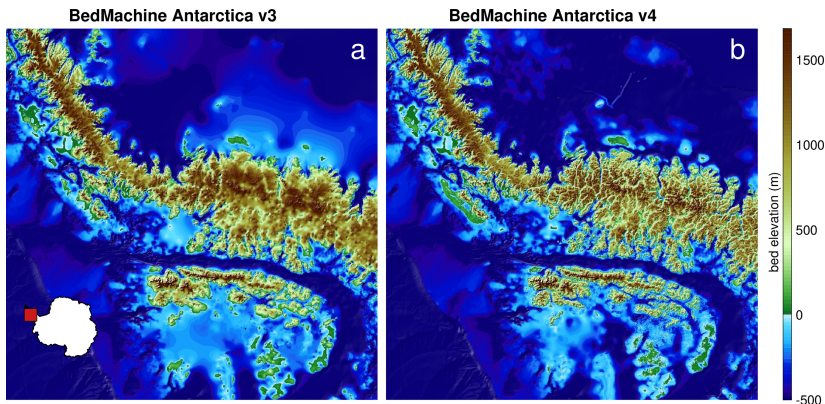


**Figure 5.** Bed topography from BedMachine Antarctica v3 (a) and v4 (b) in the area of Thwaites Glacier, West Antarctica. The white thin line shows the location of the grounding line, and the grey line shows the position of the edge of the floating ice shelf. Note that the topography under the fast-flowing region has not changed and is still based on mass conservation.

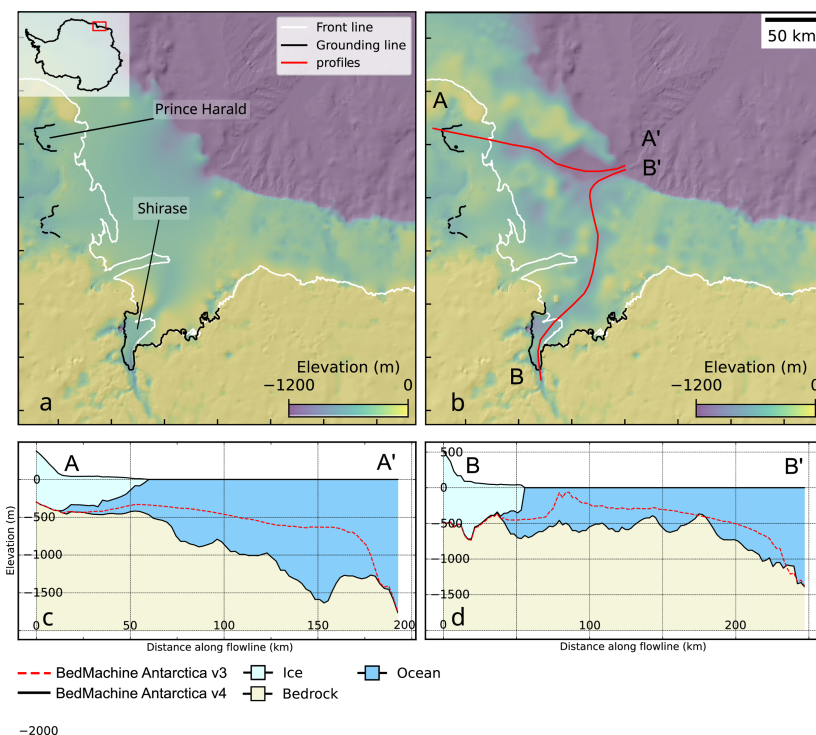
velocity product [30] to the MEaSURES phase-based velocity product [31] has significantly improved topography in the South Pole region. More details about the features captured by IFPA are discussed in [32].

For glaciers and ice caps along the periphery of the ice sheets, IceBoost reveals new fine-scale details that could not be captured by simple interpolation or traditional inversion techniques. Notably, the bed topography predicted over the Antarctic Peninsula appears more complex and rugged than the smoother BedMachine v3 (figure 6). It also reveals a network of interconnected troughs at scales of 1–3 km that result from leveraging the ice velocity field used as input. This highlights the ability of the machine learning system to generalize across different ice flow regimes, representing an improvement over traditional interpolation techniques. However, the model is sensitive to spurious input data; therefore, careful data preprocessing is required.

Finally, the bathymetry from AntGG provides critical new information that was missing in the BedMachine Antarctica v3. Figure 4, for example, shows a more complex, deeper and less smooth bathymetry under the Roi Baudouin ice shelf. By contrast, the existing bathymetry of BedMachine Antarctica is flat and featureless. Bathymetry has been inverted below the ice shelf and onto the continental shelf when lacking *in situ* data. Deep troughs are revealed, potentially linking the deep ocean to the ice shelves, as shown in figure 7 in front of the Prince Harald and Shirase ice shelves. More details captured by AntGG are discussed in [19].

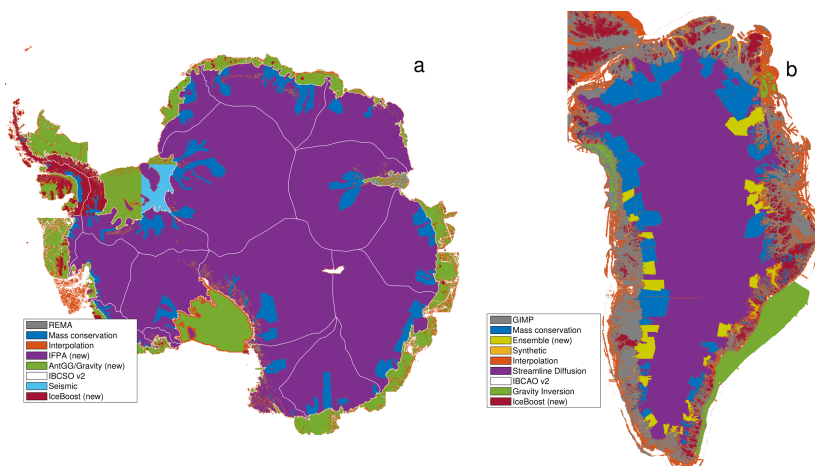


**Figure 6.** Bed topography from BedMachine Antarctica v3 (a) and BedMachine v4 (b) in the Antarctic Peninsula.



**Figure 7.** Bed topography from BedMachine Antarctica v3 (a) and v4 (b) in the area of Prince Harald and Shirase ice shelves, East Antarctica. The bed is colour coded from  $-1200$  (blue) to  $0$  m (yellow) with shaded relief. New bathymetry in (b) was computed using gravity inversions. (c) and (d) show the old and new bathymetry along the two red profiles displayed in panel (b).

Ultimately, each mapping method described here performs best in different locations, and the updated versions of BedMachine (Greenland v6 and Antarctica v4) better combine these various approaches. Figure 8 shows where the different mapping methods are applied and stitched together. We continue to use mass conservation in fast-flowing sectors of the ice sheet because this technique is consistent with the mass conservation requirements of ice sheet models. Bed topography being such an important control on ice sheet flow, we expect these new versions of BedMachine to critically improve the predictive skills of ice sheet models.



**Figure 8.** Mapping methods used in BedMachine Antarctica v4 (a) and Greenland v6 (b) showing where the four new approaches described in this paper are used. REMA is the Reference Elevation Model of Antarctica [33], IBCSO is the International Bathymetric Chart of the Southern Ocean [34], GIMP is the surface elevation from the Greenland Ice Mapping Project [22] and ‘Synthetic’ is the fjord synthetic bathymetry from [35].

## 4. Conclusion

We presented the results of applying four new approaches that help address shortcomings of the current version of BedMachine. In regions where mass conservation is applied, using an ensemble of bed estimates provides more fine-scale details in the bed topography. In the interior where simpler interpolation methods were used, IFPA captures many important details that were absent in previous versions of BedMachine. IceBoost provides an excellent alternative to existing mapping techniques for smaller ice caps and periphery glaciers, as well as the Antarctic Peninsula, which has been challenging to accurately map in the past. Finally, the new Antarctic-wide continental shelf inversion from AntGG provides a more refined representation of the ocean bathymetry around the ice sheet that captures important pathways for the intrusion of warm water into ice shelf cavities. Together, we expect these improvements to have a profound impact on future ice sheet modelling work.

**Data accessibility.** BedMachine Greenland is available at <https://nsidc.org/data/idbmg4/>, and BedMachine Antarctica is available at <https://nsidc.org/data/nsidc-0756/>.

**Declaration of AI use.** We have not used AI-assisted technologies in creating this article.

**Authors’ contributions.** M.M.: conceptualization, data curation, funding acquisition, methodology, software, supervision, visualization, writing—original draft; R.C.: data curation, visualization, writing—review and editing; N.M.: data curation, methodology, validation, writing—review and editing; R.M.: data curation, methodology, validation, writing—review and editing; H.O.: data curation, methodology, writing—review and editing; N.-J.S.: methodology, writing—review and editing; H.S.: conceptualization, methodology, writing—review and editing; M.W.: conceptualization, data curation, methodology, writing—review and editing.

All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

**Conflict of interest declaration.** We declare we have no competing interests.

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